DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY,

LUCKNOW

Proposed Evaluation Scheme & Syllabus

MBA

(Logistics and Supply Chain Management)

First Year

AS PER

AICTE MODEL CURRICULUM

(Effective from the Session: 2020-21)

MBA

(Logistics & Supply Chain Management)

Programme Objectives

Graduates with MBA (Logistics and Supply Chain Management) from Dr A P J Abdul Kalam Technical University will be able to apply the knowledge of management techniques in business environment with specific reference to Logistics and Supply Chain Management. Graduates students will have details understanding of planning, organising decision making, group dynamics, innovation, production, supply chain, operations and Technologies of today's business environment. This programme will help the graduates in designing alternatives to solve business problems utilizing quantitative analysis, critical thinking and sound ethical decision making. After completion of this programme the students will have capabilities of using research-based knowledge and methods of data collection and its interpretation to find solution to business problems. This programme is also intended to help the graduating students to apply various models and analytical techniques in managerial decision making. Within above framework and context, the Programme Objectives (PO) of MBA (Logistics and Supply Chain Management) are defined as follow.

PO 1: To understand the fundamental knowledge of management within broad framework of Supply Chain and Logistics management.

PO 2: To apply the knowledge of accounting, finance, marketing, HRM and Operations management in effective performance of Supply Chain and Logistics Management.

PO 3: To apply quantitative techniques for optimizing the decisions on Supply Chain and Logistics Management.

PO 4: Understand the operational processes at national and international levels in Supply Chain ad Logistics Management.

PO 5: To plan and organize strategies for effective Supply Chain and Logistics management.

PO 7: Able to identify the basic drivers of businesses which are impacting supply chain of the organization.

PO 8: Suggesting improvements in the production and manufacturing system on the basis of Supply Chain and Logistics

PO 9: Apply the Information technologies and latest innovations in to improve the operational performance of business.

PO 10: Communicate effectively with stakeholders with skills for report writing and manual preparations.

MBA (Logistics & Supply Chain Management) 1st Year Course Structure AICTE Model Curriculum
w.e.f. Academic Session 2020-21
Semester I

SN	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME				END SEMESTER		TOTAL	CREDIT
			L	T	P	CT	TA	PS	TOTAL	TE	PE		
1	KMLS101	Management Concepts and Organisational Behaviour	3	0	0	30	20	0	50	100	0	150	3
2	KMLS102	Managerial Economics	3	0	0	30	20	0	50	100	0	150	3
3	KMLS103	Financial Accounting and Analysis	3	1	0	30	20	0	50	100	0	150	3
4	KMLS104	Business Statistics and Data Analytics	3	1	0	30	20	0	50	100	0	150	3
5	KMLS105	Marketing Management	3	0	0	30	20	0	50	100	0	150	3
6	KMLS106	Basics of Logistics and Supply chain Management	2	0	0	15	10	0	25	50	0	75	2
7	KMLS107	Business Communication	3	1	0	30	20	0	50	100	0	150	3
Lab/Practical													
8	KMLS151	IT Skills Lab -1	0	0	3	-	-	50	50	-	100	150	3
9	KMLS152	Mini Project -1	0	0	2	-	-	25	25		50	75	3
												1200	26

MBA
(Logistics & Supply Chain Management)

Semester II

SN	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME				END SEMESTER		TOTAL	CREDIT
			L	T	P	CT	TA	PS	Total	TE	PE		
1	KMLS201	Business Environment and Legal Framework	3	0	0	30	20	0	50	100	0	150	3
2	KMLS202	Human Resource Management	3	0	0	30	20	0	50	100	0	150	3
3	KMLS203	Business Research Methods	3	0	0	30	20	0	50	100	0	150	3
4	KMLS204	Financial Management and Corporate Finance	3	1	0	30	20	0	50	100	0	150	3
5	KMLS205	Operations and Green Supply Chain Management	3	1	0	30	20	0	50	100	0	150	3
6	KMLS206	Quantitative Techniques for Managers	3	1	0	30	20	0	50	100	0	150	3
7	KMLS207	Procurement, Storage and Warehouse Management	3	0	0	30	20	0	50	100	0	150	3
8	KMLS208	Port and Airport Management for Logistics	2	0	0	15	10	0	25	50	0	75	2
Labs/Practical													
9	KMLS251	IT Skills Lab-2	-	-	2	0	0	25	25	0	0	25	1
10	KMLS252	Mini Project -2	-	-	2	-	-	25	25	-	25	50	2
												1200	26

I Semester

MANAGEMENT CONCEPTS AND ORGANISATIONAL BEHAVIOUR

Course Credit: 3 Contact Hours: 40

Course Objectives:

1. To provide basic understandings of management processes
2. To help the students understand the concepts of organizational behaviour
3. To apply the concepts of management and organizational behaviors in real world situations
4. Familiarizing the students with the contemporary issues in management.
5. Developing managerial and leadership skills among students

UNIT I (8 Lectures)

Fundamentals of Management: Management practices from past to present, Different levels of management, Managerial skills and Managerial Functions , Case Studies

Planning- Objective of planning, Planning process, Types of planning, Types of plans, Management by Objective, Decision-making- types, process & techniques,. Case Studies

UNIT-II (8 Lectures)

Organising & Staffing- Types of organization, Organization structure and decentralization of authority, Meaning of staffing, Recruitment, selection & placement, Training & development..

Directing & Controlling- Principle of directing, Essence of coordination, Different control techniques, Management by exception. Case Studies

UNIT III(8 Lectures)

Fundamentals of individual behavior, Personality, types of personality, Personal effectiveness, meaning of Attitudes, Types, Components, attitude formation and attitude change. Meaning & Type of Group Behaviour, Interpersonal skills, Transactional Analysis, Johari Window,

UNIT IV (8 Lectures)

Motivation:, Theory of Motivation: Maslow's, Herzberg's, McClelland, Contemporary theories of Motivation: Self Determination Theory, Self Efficacy Theory, Vroom's Expectancy Theory, Equity Theory, Reinforcement Theory, Meaning of Perception, process, behavioral applications of perception. Case Studies

UNIT V: (8 Lectures)

Leadership: What is leadership, types of leaders and leadership styles, traits and qualities of effective leader, trait theory, LSM – Leadership Situational Model, Team Building, Tuckman Model of Team Development. Organizational Change: Meaning of organizational change approaches to managing organizational change, creating a culture for change, implementing the change, Kurt Lewin Model of change. Case Studies

COURSE OUTCOME

Course Outcomes	Bloom's taxonomy
CO 1: Developing understanding of managerial practices and their perspectives.	• Knowledge (K 2) • Remembering (K1)
CO2: Understanding and Applying the concepts of organizational behaviour	• Knowledge (K 2) • Applying (K 4)
CO 3: Applying the concepts of management and analyze organizational behaviors in real world situations	• Applying (K 4) • Analyzing (K 5)
CO 4: Comprehend and practice contemporary issues in management.	• Comprehending (K 3)
CO 5: Applying managerial and leadership skills among students	• Applying (K4)

Suggested Readings

1. Koontz Harold &Weihrich Heinz – Essentials of management (Tata McGraw Hill, 5th Edition,2008)
2. L. M. Prasad- Principles and Practices of Management, Sulatn Chand & Sons, 7th edition, 2007.
3. Stephen P. Robbins, ―Organizational Behaviour‖, 12th Edition, Prentice Hall
4. Dr. Premvir Kapoor, Principles and Practices of Management, Khanna Publishing House, Delhi
5. Robbins & Coulter - Management (Prentice Hall of India, 9th Edition)
6. Principles of Management, George R. Terry & S.G. Franklin, AITBS, Delhi.
7. N M Khandelwal- Indian Ethos & Values for Management- Himalyan Publishing
8. Fred Luthans, ―Organizational Behaviour‖, 12th Edition, McGraw Hill International Edition
9. Aswathappa K, ―Organizational Behaviour (Text, Cases and Games)‖, Himalaya Publication
10. UdaiPareek, ―Organizational Behavior‖, Oxford University Press

MANAGERIAL ECONOMICS

Course Credit: 3 Contact Hours: 40

Course Objective:

1. To understand the importance of Managerial Economics in management and businesses
2. To apply the principles of managerial economics in achieving business objectives
3. Be equipped with the tools necessary in forecasting product demand
4. Understand and be able to apply latest pricing strategies
5. Understand and analyze the macro environment affecting the business decision making.

UNIT –I (6 Hours)

Basic Concepts and principles: Definition, Nature and Scope of Economics-Micro Economics and Macro Economics, Managerial Economics and its relevance in business decisions. Fundamental Principles of Managerial Economics - Incremental Principle, Marginal Principle, Opportunity Cost Principle, Discounting Principle, Concept of Time Perspective, Equi-Marginal Principle, Utility Analysis, Cardinal Utility and Ordinal Utility. Case Studies

UNIT –II (8Hours)

Demand and Supply Analysis: Theory of Demand, Types of Demand. Determinants of demand, Demand Function, Demand Schedule, Demand curve, Law of Demand, Exceptions to the law of Demand, Shifts in demand curve, Elasticity of Demand and its measurement. Price Elasticity, Income Elasticity, Arc Elasticity. Cross Elasticity and Advertising Elasticity. Uses of Elasticity of Demand for managerial decision making, Demand forecasting meaning, significance and methods.(numerical Exercises) Case Studies

Supply Analysis; Law of Supply, Supply Elasticity; Analysis and its uses for managerial decision making.

Price of a Product under demand and supply forces . Case Studies

UNIT –III (10Hours)

Production and cost Analysis: Production concepts & analysis; Production function, Types of production function, Laws of production: Law of diminishing returns, Law of returns to scale.

Cost concept and analysis: Cost, Types of costs, Cost output relationship in the short-run. Cost output relationship in the Long-run. Estimation of revenue. Average Revenue, Marginal Revenue . Case Studies

UNIT –IV (10Hours)

Market structures: Perfect and Imperfect Market Structures, Perfect Competition, features, determination of price under perfect competition. Monopoly: Feature, pricing under monopoly, Price Discrimination. Monopolistic: Features, pricing under monopolistic competition, product differentiation. Oligopoly: Features, kinked demand curve, cartels, price leadership. Case Studies

UNIT –V (6Hrs)

National Income; Concepts and various methods of its measurement, Circular flows in 2 sector, 3 sector, 4 sector economies, Inflation, types and causes, Business Cycle & its phases.

Course Outcomes:

Course Outcomes	**Bloom's taxonomy**
CO1: Students will be able to remember the concepts of micro economics and also able to understand the various micro economic principles to make effective economic decisions under conditions of risk and uncertainty.	• Knowledge (K 2) • Remembering (k1)
CO2: The students would be able to understand the law of demand & supply & their elasticities , evaluate & analyse these concepts and apply them in various changing situations in industry . Students would be able to apply various techniques to forecast demand for better utilization of resources.	• Knowledge (K 2) • Applying (K 4) • Synthesizing (K6) • Evaluating (K7)
CO3: The students would be able to understand the production concept and how the production output changes with the change in inputs and able to analyse the effect of cost to business and their relation to analyze the volatility in the business world	• Comprehending (K 3) • Applying (K 4) • Analyzing (K 5) • Evaluating (K7)
CO4: The students would be able to understand & evaluate the different market structure and their different equilibriums for industry as well as for consumers for the survival in the industry by the application of various pricing strategic	• Applying (K 4) • Analyzing (K 5) • Synthesizing (K6)
CO5: The students would be able to analyse the macroeconomic concepts & their relation to micro economic concept & how they affect the business & economy.	• Knowledge (K 2) • Comprehending (K 3)

Suggested Readings

1. Managerial Economics ,D.N.Dwivedi,Vikas Publication, 7th Ed
2. Managerial Economics, GEETIKA, McGraw-Hill Education 2nd Ed.
3. Managerial Economics: Concepts and Applications (SIE), THOMAS& MAURICE, McGraw-Hill Education, 9th Ed
4. Managerial Economics, H.L Ahuja, S.Chand, 8th Ed
5. Managerial Economics – Theory and Applications, Dr.D.M.Mithani, Himalaya Publications, 7th Ed.
6. Sociology & Economics for Engineers, Dr. Premvir Kapoor, Khanna Publishing House

FINANCIAL ACCOUNTING AND ANALYSIS

Course Credit: 3 Contact Hours: 40

Course Objectives:

1) To understand the fundamentals, basic theory and concepts of financial accounting.
2) To have a knowledge about various Accounting Standards used in preparation of financial statements.
3) To have an understanding of preparation and presentation of financial statements.
4) To acquire knowledge about various techniques used for analysing financial statements with its application.
5) To enable students acquainted with current trends and social responsibility accounting.

UNIT I (6Hrs)

Meaning and Scope of Accounting: Evolution and Users of Accounting, Basic Accounting terminologies, Principles of Accounting, Accounting Concepts & Conventions, Accounting Equation, Deprecation Accounting.

UNIT II (6Hrs)

Mechanics of Accounting: Accounting Standards and IFRS: International Accounting Principles and Standards; Matching of Indian Accounting Standards with International Accounting Standards, Double entry system of Accounting, journalizing of transactions; Ledger posting and Trial Balance.

UNIT III (12 Hrs)

Presentation of Financial Statement: Preparation of final accounts (Profit & Loss Account and Balance Sheet) according to companies act 2013 (vertical format), Excel Application to make Balance sheet, Case studies and Workshops, Preparation of Cash Flow Statement and its analysis.

UNIT IV (10 Hrs)

Analysis of financial statement: Ratio Analysis- Solvency ratios, Profitability ratios, activity ratios, liquidity ratios, Market capitalization ratios; leverage Ratio, Detailed Analysis using excel application.

UNIT V (6 Hrs)

Financial Statement Analysis and Recent Types of Accounting: Common Size Statement; Comparative Balance Sheet and Trend Analysis of manufacturing, Service & banking organizations, Case Study and Workshops in analysing Balance sheet. Human Resource Accounting, Forensic Accounting, Accounting for corporate social responsibility.

Course Outcome:

After successful completion of this course students will be able to

S.No	Course Outcome	Bloom's Taxonomy
1	CO1.Understand and apply accounting concepts, principles and conventions for their routine monetary transaction.	Knowledge (K2)/ Comprehending (K 3)
2	CO2. Understand about IFRS, Ind AS and IAS for preparation and reporting of financial statements.	Knowledge (K2) Synthesizing (K6)/
3	CO3. Create and prepare financial statements and Cash flow in accordance with Generally Accepted Accounting Principles	Remembering (k1)
4	CO4. Analyse, interpret and communicate the information contained in basic financial statements and explain the limitations of such statements.	Analysing (K 4) / Evaluating (K7))
5	CO5. Recognising various types of accounting and utilize the technology and social responsibility in facilitating and enhancing accounting and financial reporting processes	Knowledge (K2) Applying (K 4)

Suggested Readings

1. Maheshwari S.N &Maheshwari S K – A text book of Accounting for Management (Vikas, 10th Edition)
2. Essentials of Financial Accounting (based on IFRS), Bhattacharya (PHI,3rd Ed)
3. Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
4. PC Tulsian- Financial Accounting (Pearson, 2016)
5. Dhamija - Financial Accounting for managers: (Prentice Hall, 2nd Edition).
6. Narayanswami - Financial Accounting: A Managerial Perspective (PHI,5th Ed)
7. DhaneshkKhatri- Financial Accounting (TMH,2015)
8. Ambrish Gupta - Financial Accounting: A Managerial Perspective (Prentice Hall, 4th Edition)
9. Ramchandran&Kakani - Financial Accounting for Management (TMH, 2nd Edition).
10. Mukherjee - Financial Accounting for Management (TMH, 2nd Edition).

BUSINESS STATISTICS & ANALYTICS

Course Credit: 3 Contact Hour 40 hours

Course Objectives

1. Understand the different basic concept / fundamentals of business statistics.
2. Understand the importance of measures of Descriptive statistics which includes measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.
3. Understand the concept of Probability and its usage in various business applications.
4. Understand the Hypothesis Testing concepts and use inferential statistics- t, F, Z Test and Chi Square Test
5. Understand the practical application of Descriptive and Inferential Statistics concepts and their uses for Business Analytics.

Unit I (10 Sessions): Descriptive Statistics

Meaning, Scope, types, functions and limitations of statistics, Measures of Central tendency – Mean, Median, Mode, Quartiles, Measures of Dispersion – Range, Inter quartile range, Mean deviation, Standard deviation, Variance, Coefficient of Variation, Skewness and Kurtosis.

Unit II (8 Sessions):Time Series & Index Number

Time series analysis: Concept, Additive and Multiplicative models, Components of time series, Trend analysis: Least Square method - Linear and Non- Linear equations, Applications in business decision-making.

Index Numbers:- Meaning , Types of index numbers, uses of index numbers, Construction of Price, Quantity and Volume indices:- Fixed base and Chain base methods.

Unit III (6 Sessions): Correlation & Regression Analysis

Correlation Analysis: Rank Method & Karl Pearson's Coefficient of Correlation and Properties of Correlation.

Regression Analysis: Fitting of a Regression Line and Interpretation of Results, Properties of Regression Coefficients and Relationship between Regression and Correlation.

Unit IV (8 Sessions): Probability Thoery & Distribution

Probability: Theory of Probability, Addition and Multiplication Law, Baye's Theorem

Probability Theoretical Distributions: Concept and application of Binomial; Poisson and Normal distributions.

Unit V (8 Sessions) *Hypothesis Testing& Business Analytics*

Hypothesis Testing: Null and Alternative Hypotheses; Type I and Type II errors; Testing of Hypothesis: Large Sample Tests, Small Sample test, (t, F, Z Test and Chi Square Test)

Concept of Business Analytics- Meaning types and application of Business Analytics, Use of Spread Sheet to anlayze data-Descriptive analytics and Predictive analytics.

Course Outcomes

Course Outcome	Blooms Taxanomy
CO1. Gaining Knowledge of basic concept / fundamentals of business statistics.	• Knowledge (K 2)

CO2. To compute various measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.	• Remembering (K1) • Applying (K 4)
CO3. Evaluating basic concepts of probability and perform probability theoretical distributions	• Comprehending (K 3) • Applying (K 4)
CO4. To apply Hypothesis Testing concepts and able to apply inferential statistics- t, F, Z Test and Chi Square Test	• Analyzing (K 5) • Synthesizing (K6)
CO5. To perform practical application by taking managerial decision and evaluating the Concept of Business Analytics.	• Evaluating (K7) • Applying (K 4)

Suggested Readings

1. G C Beri – Business Statistics, 3rd ed, TATA McGrawHill.
2. Chandrasekaran & Umaparvathi-Statistics for Managers, 1st edition, PHI Learning
3. Davis , Pecar – Business Statistics using Excel, Oxford
4. Ken Black – Business Statistics, 5th ed., Wiley India
5. Levin and Rubin – statistics for Management, 7th ed., Pearson
6. Lind, Marchal, Wathen – Staistical techniques in business and economics, 13th ed, McGrawHill
7. Newbold, Carlson, Thorne – Statistics for Business and Economics, 6th ed., Pearson
8. S. C.Gupta – Fundamentals of Statistics, Himalaya Publishing
9. Walpole – Probability and Statistics for Scientists and Engineers, 8th ed., Pearson

MARKETING MANAGEMENT

COURSE CREDIT: 3 Contact Hour 40 Course Objectives

1. Assess market opportunities by analyzing customers, competitors, collaborators, context, and the strengths and weaknesses of a company.
2. Understand consumers' requirements and their behaviors.
3. Develop effective marketing strategies to achieve organizational objectives.
4. Communicate and defend your recommendations and critically examine and build upon the recommendations of your classmates both quantitatively and qualitatively.
5. Develop the understanding the current global and digital aspect of marketing.

Unit 1(6 hours)

Introduction: Nature and scope of marketing, Various marketing orientations, Need, Want, Demand, Elements of Marketing mix, customer value and the value delivery process.

Understanding Consumer Behavior: Buying motives, factors influencing buying behavior, buying habits, stages in consumer buying decision process, types of consumer buying decisions.

Unit 2 (8 hours)

Market segmentation, Targeting and Positioning: Meaning, Factors influencing segmentation, Market Aggregation, Basis for segmentation, Segmentation of Consumer. Targeting: Meaning, Basis for identifying target customers, Target Market Strategies. Positioning: Meaning, product differentiation strategies, tasks involved in positioning. Branding: Concept of Branding, Brand Types, Brand equity, Branding Positioning.

Unit 3 (8 hours)

Product Decisions: Concept, product hierarchy, new product development, diffusion process, Product Life cycle, Product mix strategies. Packaging / Labeling: Packaging as a marketing tool, requirement of good packaging, Role of labeling in packaging. Pricing Decisions: Pricing concepts for establishing value, Pricing Strategies-Value based, Cost based, Market based, Competitor based, New product pricing – Price Skimming & Penetration pricing

Unit 4 (8 hours)

Place Decision: Meaning, Purpose, Channel alternatives, Factors affecting channel choice, Channel design and Channel management decisions, Channel conflict, Retailing & Types of Retailers. Advertising: Advertising Objectives, Advertising Budget, Advertising Copy, AIDA model, Public Relation: Meaning, Objectives, Types, and Functions of Public Relations. Sales Promotion: Sales Promotion Mix, Kinds of promotion, Tools and Techniques of sales promotion, Push-pull strategies of promotion, Personal Selling: Concept, Features, Functions, Steps/process involved in Personal Selling, Direct Marketing: Meaning, Features, Functions, Growth and benefits of direct marketing, different forms.

Unit 5 (6 hours)

CRM: Meaning, Relationship Marketing Vs. Relationship Management, Types of Relationship Management, Significance of Customer Relationship Management. Global Marketing: current scenario, Global Marketing environment, Entry strategies, Global P's of Marketing., Recent trends and Innovation in Marketing- Green Marketing, Agile Marketing

Course Outcomes: Upon the successful completion of this course, the student will be able to:

S.No	Course Outcome	Bloom's taxonomy
1	CO1. Remember and Comprehend basic marketing concepts.	• Remembering (k1) • Knowledge (K 2)
2	CO2. Understand marketing Insights on application of basic marketing concepts.	• Synthesizing (K6) • Comprehending(K3)
3	CO3. Able to Apply and develop Marketing Strategies and Plans	• Applying (K 4)
4	CO4. Understand and Analyzing Business/ Consumer Markets and ability Identify & evaluate Market Segments and Targeting	• Analyzing (K 5)
5	CO5. Develop skills to understand the current global and digital aspect of marketing.	• Evaluating (K7)

Recommended Text Books :

1. Marketing Management: A South Asian Perspective - Kotler, Keller, Kevin 15/e, Pearson Education, 2016.
2. Marketing Management - Ramaswamy V. S. & Namakumari S, 6/e, Sage Publication India Pvt Ltd., 2018.
3. Marketing Management - Tapan Panda, 5/e, Excel Publication, 2007.
4. Fundamentals of Marketing Management - Etzel M. J, B J Walker & William J. Stanton, 14/e, McGrawHill Education Publishers, 2015.
5. Marketing: Asian EditionPaul Bainies, Chris Fill Kelly Page third edition, Oxford.

BUSINESS COMMUNICATION

Course Credits: 3 Contact Hours: 40

Course Objectives

1. To understand business communication strategies and principles for effective communication in domestic and international business situations.
2. To understand and appropriately apply modes of expression, i.e., descriptive, expositive, narrative, scientific, and self-expressive, in written, visual, and oral communication.
3. To develop the ability to research and write a documented paper and/or to give an oral presentation.
4. To develop the ability to communicate via electronic mail, Internet, and other technologies for presenting business messages.
5. To understand and apply basic principles of critical thinking, problem solving, and technical proficiency in the development of exposition and argument.

UNIT I : (8 Hours)

Introduction: Role of communication – defining and classifying communication – purpose of communication – process of communication –characteristics of successful communication – importance of communication in management – communication structure in organization – communication in crisis barriers to communication. Case Studies

UNITII: (8 Hours)

Oral communication: What is oral Communication – principles of successful oral communication – what is conversation control – reflection and empathy: two sides of effective oral communication – effective listening – non – verbal communication. Written communication: Purpose of writing – clarity in writing – principles of effective writing – approaching the writing process systematically: The 3X3 writing process for business communication: Pre writing – Writing – Revising – Specific writing features – coherence – electronic writing process.

UNITIII: (8 Hours)

Business letters and reports: Introduction to business letters – writing routine and persuasive letters – positive and negative messages- writing memos – what is a report purpose, kinds and objectives of report writing. Presentation skills: What is a presentation – elements of presentation – designing a presentation. Advanced visual support for business presentation types of visual aid

UNITIV: (8 Hours)

Employment communication: Introduction – writing CVs – Group discussions – interview skills Impact of Technological Advancement on Business Communication networks – Intranet – Internet – e mails – SMS – teleconferencing – video conferencing. Case Studies

UNITV : (8 Hours)

Group communication: Meetings – Planning meetings – objectives – participants – timing – venue of meetings – leading meetings. Media management – the press release press conference – media interviews Seminars – workshop – conferences. Business etiquettes. Case Studies

Course Outcomes

Upon successful completion of this course, the student should be able to:

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Apply business communication strategies and principles to prepare effective communication for domestic and international business situations.	Applying (K4)
2	CO2. Analyse ethical, legal, cultural, and global issues affecting business Communication.	Analyse (K5)
3	CO3. Develop an understanding of appropriate organizational formats and channels used in business communications	Knowledge (K2)
4	CO4. Gaining an understanding of emerging electronic modes of communication.	Comprehending(K3)
5.	CO5. Developing effective verbal and non verbal communication skills.	Remembering(K1)/ Applying (K4)

BASICS OF SUPPLY CHAIN & LOGISTICS MANAGEMENT

Course Credits 3 Contact Hours 30

Course Objectives:

This course is intended to provide an understanding of the components and processes of supply chain and logistics management as well as the performance drivers of supply chain. It is also intended to help the students to learn about logistics, transportation, warehousing and outsourcing decisions.

Unit 1 (6 Hours)

Supply Chain Concepts: Objectives of a Supply Chain, Stages of Supply chain, Value Chain Process, Cycle view of Supply Chain Process, Key issues in SCM, logistics & SCM, Supply Chain Drivers and obstacles, Supply chain strategies, strategic fit, Best practices in SCM, Obstacles of streamlined SCM.

Unit 2 (6 Hours)

Logistics :Evolution, Objectives, Components and Functions of Logistics Management, Distribution related Issues and Challenges; Gaining competitive advantage through Logistics Management, Transportation- Functions, Costs, and Mode; Network and Decision, Containerization, Cross docking.

Unit 3 (8 Hours)

Supply Chain Performance: Bullwhip effect and reduction, Performance measurement: Dimension, Tools of performance measurement, SCOR Model. Demand chain management, Global Supply chain- Challenges in establishing Global Supply Chain, Factors that influences designing Global Supply Chain Network.

Unit 4 (12 Hours)

Warehousing: Concept and types, Warehousing strategy, Warehouse facility location & network design, Reverse logistics, Outsourcing- Nature and concept, Strategic decision to Outsourcing, Third party logistics(3PL), Fourth party logistics(4PL).

Supply Chain and CRM- Linkage, IT infrastructure used for Supply Chain and CRM, Functional components for CRM, Green supply chain management, Supply Chain sustainability.

Suggested Readings:

1. Chopra, Sunil, Meindl, Peter and Kalra, D. V.; Supply Chain Management: Strategy, Planning
1. and Operation; Pearson Education
2. Altekar, Rahul V.; Supply Chain Management: Concepts and Cases; PHI Learning Reference
3. Books
4. Ballou, Ronald H.; Supply Chain Management; Pearson Education
5. Sahay, B.S.; Supply Chain Management; Macmillan
6. Ballou, R.H. Business Logistics Management. Prentice-Hall Inc.
7. Bowersox D.J. ,Closs D.J. , Logistical Management, McGraw-Hill, 1996

IT SKILLS LAB-1

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To provide knowledge about the functioning of computers and its uses for managers
2. To provide hands on learning on Internet and its applications
3. To provide hands on learning on Word processing software
4. To provide hands on learning of applications on Spreadsheet software
5. To provide hands on learning on Presentation software

UNIT I (05 hours) Conceptual Framework

Hardware: (a) Input devices - keyboard, printing devices, voice speech devices, scanner, MICR, OMR, Bar code reader, digital camera etc. (b) Output devices - Visual Display UNIT, printers, plotters (c) Storage Devices – Magnetic storage devices, Optical storage devices, Flash Memory.

Software: Types of software with examples; Introduction to languages, compiler, interpreter and Assembler, Operating System Functions, Types and Classification, Elements of GUI based operating system. Network and Internet: Types of computer networks (LAN, WAN and MAN), Netiquettes, Basic services over Internet like WWW, FTP, Telnet, Gopher, URL, Domain names, Web Browsers, Multimedia and its applications: Concepts of Text, Graphics, Animation, Audio, Images, Video. Multimedia Application in Education, Entertainment, Marketing. Names of common multimedia file formats,

UNIT II : Windows and Users Interface (Lab Work)- 7 hours

Windows operating System: Introduction and characteristics, Elements of GUI. Using Mouse, My Computer Icon, The Recycle Bin, Status Bar, Start and Menu & Menu-selection, Running an Application, Windows Explorer: Viewing of File, Folders and Directories Creating and Renaming of files and folders Opening and closing of different Windows, Windows Setting: Control Panels, Wall paper and Screen Savers Setting the date and Sound. Concept of menu, Using Help, Using right Button of the Mouse, Creating Short cuts, Basics of Window Setup, Notepad, Window Accessories

UNIT III: Word Processor Software (Lab Work) – 8 hours

Word processing concepts: Opening, Saving, Closing the file, Opening an existing document, Selecting text, Editing text, Finding and replacing text, printing documents, Creating and Printing Merged Documents, Character and Paragraph Formatting, Page Design and Layout. Editing and Profiling Tools: Checking and correcting spellings. Using Graphics, Tables, Charts, Document Templates and Wizards.

UNIT IV: Spreadsheet Software (Lab Work) – 10 hours

Spreadsheet Package Spreadsheet: Concept and Working Interface, Creating, Saving and Editing a Workbook, Inserting, Deleting Work Sheets, entering data in a cell / formula Copying and Moving from selected cells, handling operators in Formulae. Functions in Spreadsheet: Mathematical, Logical, statistical, text, financial, Date and Time functions, Using Function Wizard. Formatting a Worksheet and Cell: changing data alignment, changing date, number, character or currency format, changing font, adding borders and colors. Printing worksheets, Charts and Graphs – Creating, Previewing, and Modifying Charts. Integrating word processor, spread sheets, web pages.

UNIT V: Presentation Software (lab Work) – 8 hours

Interface of the Presentation Package: Creating, Opening and Saving Presentations. Professional Look of the Presentation: Working in different Design & Views, Working with Slides. Formatting and Editing: Text, Image and Paragraph formatting, Checking Spelling and Correcting Typing Mistakes, Making Notes Pages and Handouts, Drawing and Working with Objects, Adding Clip Art and other pictures, Designing Slide Shows, Running and Controlling a Slide Show, Printing Presentations.

Course Outcomes

Upon successful completion of this course, the student should be able to:

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Gain in depth knowledge about the functioning of computers and its uses for managers	Knowledge (K2)
2	CO2. Learn to use Internet and its applications	Applying (K4)
3	CO3. Understand and implement Word processing software	Synthesizing (K6) Applying (K4)
4	CO4. Learn applications on Spread sheet softwares	Knowledge (K2)
5	CO5. Analyse and learn Presentation software	Analyse (K5)

Suggested Readings

1. Nasib Singh Gill – Handbook of Computer Fundamentals, Khanna Publishing House, Delhi
2. Shrivastava-Fundamental of Computer& Information Systems (Wiley Dreamtech)
3. Leon A and Leon M - Introduction to Computers (Vikas, 1st Edition).
4. ITL ESL – Introduction to Information Technology (Pearson, 2nd Edition).
6. Introduction to Computers, Norton P. (TATA McGraw Hill)
7. Leon - Fundamentals of Information Technology, (Vikas)
8. Satish Jain-BPB's Computer Course Windows 10 with MS Office 2016 (BPB)
9. Linda Foulkes- Learn Microsoft Office 2019: A comprehensive guide to getting started with Word, PowerPoint, Excel, Access, and Outlook (Packt Publishing Limited)

MINI PROJECT -1

Course Credit -2

Course Objective-

1. To develop an innovative idea for product or services in form of a project report.
2. To understand importance and relevance of innovative idea, its feasibilities and detail descriptions.

Project/Practical work / Seminar

In first semester, the students are required to develop an innovative idea for product or services and a project report to be prepared on that idea under the guidance of faculty member. Report will be prepared individually and this report will consist of importance and relevance of innovative idea, its feasibilities and detail descriptions. The report will be evaluated by one external examiner appointed by university. Student has to present his output through a seminar.

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Gain in depth knowledge on innovative idea for product or services in form of a project report.	Knowledge (K2)
2	CO2. To apply innovative idea, its feasibilities and detail descriptions.	Applying (K4)

Semester II

Business Environment & Legal Aspect of Business

Course Credit: 3 Contact Hours: 40

Course Objectives:

1. The basic objective of the course is to develop understanding and provide knowledge about business environment to the management students.

2. To promote basic understanding on the concepts of Business Environment and international business environment.

3. To provide basic understanding of law of contract

4. To impart basic understanding of provisions of Companies Act concerning incorporation and regulation of business organizations.

5. To appraise the students on the leading practical application oriented case studies – relevant and updated and analyzing case laws in arriving at conclusions facilitating business decisions.

Unit I - (10Hrs)
Introduction to Micro Environment –
Meaning of Business & Business Environment,, Types of Business Organizations , SWOT analysis , Types of Environment-Internal to the Enterprise (Value System, Management Structure and Nature, Human Resource, Company Image and Brand Value, Physical Assets, Facilities, Research & Development, Intangibles, Competitive Advantage), External to the Enterprise , Micro- Suppliers, Customers, Market Intermediaries; Macro- Demography, Natural, Legal & Political, Technological,) Michael Porter's Five Forces Analysis, Competitive Strategies
Unit II - (6 Hrs)
Macro Cont: Economic, Socio-Cultural, Competitive & International Environment –
Economy, Competition, Socio-cultural and International); Business Environment with reference to Global Integration; Comparative Analysis of Business Environment: India and Other Countries , Factors affecting international business environment, Business Policy : LPG model & International forces in business.
UNIT- III (8 hrs)

Law of Contract: Definition, essentials and types of contracts, offer definition and essentials, acceptance – definition and essentials, consideration – definition and essentials, exceptions to the rule, no consideration, no contract, doctrine of privity of contract, capacity of parties, free consent, quasi contract, legality of object, performance of contract, termination of contract, remedies for breach of contract.
Sale of Goods Act: Essentials, sale v/s agreement to sell. Condition v/s warranties, rights of unpaid seller

UNIT IV (8hrs)

Companies Act Definition, characteristics and kinds of companies, steps in formation of company. Memorandum of Association, Articles of Association, prospectus. Directors: appointment, power, duties and liabilities, meeting and resolutions: types of meetings. Auditor: appointment, rights and liabilities, modes of winding up of a company.

UNITV (8 hrs)

Consumer Protection Act: Definitions - Aims and objectives, Consumer protection councils, Redressal agencies and penalties for violation.

The Information Technology Act: Definition, Digital Signature, Electronic Governance, Attribution, Acknowledgment and Dispatch of Electronic Records, Sense Electronic Records and Sense Digital Signatures, Regulation of Certifying Authorities, Digital Signature Certificates, Duties of Subscribers, Penalties and Offences.

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1) Develop understanding and fundamental knowledge about business environment	Remembering (k1) Knowledge (K 2)
2	CO2) Develop understanding on the concepts of Business Environment and international business environment.	K2 Knowledge
3	CO3) Develop basic understanding of law of contract	K2 Knowledge
4	CO4) understanding of provisions of Companies Act concerning incorporation and regulation of business organizations	K2 Knowledge
5	CO5) Able to analyze case laws in arriving at conclusions facilitating business decisions.	K4 Applying K5 Analysing

Suggested Readings

i. Business Environment ---Francis Cherunilam, Himalaya Publishing House
ii. Business Environment: Test and Cases , PAUL, Mc Graw Hill Education , 3rd Ed.
iii. V. Neelamegam – Business Envirnoment (VrindaPublications , 2nd Edition)
iv. Shaikh & Saleem - Business Environment (Pearson, 2nd Edition)
v. International Business Environment—Ian Brooks, Jamie Weatherstom and GrahmWilkinson
vi. Kuchhal M.C. - Business Law (Vikas Publication)
vii. Gulshan S.S. - Business Law Including Company Law (Excel Books)
viii. N D Kapoor – Elements of Mercantile Law – Sultan Chand-2014.

HUMAN RESOURCE MANAGEMENT

Course Credit: 3 Contact Hours: 40

Course Objectives: In this course the students will learn the basic concepts and frameworks of Human Resource Management (HRM) and understand the role that HRM has to play in effective business administration. It will provide an insight as to how to use Human Resource as a tool to implement strategies.

UNIT I: (7 Hours)

Essentials of HRM: Functions of HRM, HRM vs.HRD, Strategic HRM: Meaning and Roles in Strategy formulation and implementation, Barriers to strategic HRM, Linking HR strategy with business strategy, Roles of HR Manager, roles of HR in merger and acquisitions, Technology & HR and changing roles of HR due to technology, HRM linkage with TQM & productivity. Case Studies

UNIT II: (8 Hours)

Human Resource Planning and Employee Hiring : Meaning of job Analysis, job design, Human Resource Planning, methods demand forecasting for manpower planning, factors influencing HRP, Employee hiring- methods of Recruitment, Employee selection, process of employee selection, recent trends in recruitment. Case Studies

UNIT III: (8 Hours)

Employee Training & Development: Meaning importance of Training, types and methods and types of training, career planning, promotion, transfer, demotion and separation, Performance Appraisal: Meaning and types of appraisal, Job Evaluation: Meaning and methods of job evaluation. Case Studies

UNIT IV: (9 Hours)

Compensation Management and Employee Relations: Introduction to compensation management, Components and structure of employee compensation, Factors affecting employee compensation, Employee incentive schemes, and recent trends in compensations management, Meaning of employee relation and industrial relations. Case Studies

UNIT V: (8 Hours)

Employee Safety/ Health and International Human Resource Management: Needs and leagal provision of employee health, measures to promote employee health , purpose of employee safety, accidents: causes & prevention, effective safety management ,& legal provisos. basic principles governing International Human Resource Case Studies

COURSE OUTCOME

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Synthesize the role of human resources management as it supports the success of the organization including the effective development of human capital as an agent for organizational change.	K6 Synthesizing
2	CO2. Demonstrate knowledge of laws that impact behaviour in relationships between employers and employees that ultimately impact the goals and strategies of the organization.	K2 Knowledge
3	CO3. Understand the role of employee benefits and compensation as a critical component of employee performance, productivity and organizational effectiveness.	K3 Comprehending
4	CO4.Show evidence of the ability to analyze, manage and problem solve to deal with the challenges and complexities of the practice of collective bargaining.	K5Analysing
5	CO5. Demonstrate knowledge of practical application of training and employee development as it impacts organizational strategy and competitive advantage.	K2 Knowledge K4 Applying

Suggested Readings

1. V.S.P.Rao, Human Resource Management (Text and Cases) Himalaya Publications, Thirteenth Edition.

2. Durai Praveen, Human Resource Management Pearson Publication, 2nd Edition.

3. Gary Dessler and BijuVarkkeyHuman Resource Management, Person Publication, 2013, 14th Edition.

4. SeemaSanghi, Human Resource Management, VikasPubllications, 2014, 5th Edition.

5. K. Aswathappa, Human Resource Management, McGraw Hill Education, 2013, 7th Edition.

BUSINESS RESEARCH METHODS

Course Credit: 3 Contact Hours:40

Course objectives

1. Understand the concept / fundamentals of research and their types.
2. Understand the practical application of various research techniques.
3. Understand the importance of scaling & measurement techniques and sampling techniques
4. Understand the importance of coding, editing, tabulation and analysis in doing research.
5. Understanding and applying the concept of statistical analysis which includes ANOVA technique and technique of report writing.

Unit 1 (8 Sessions)

Research: – Definition, Meaning, Importance types and Qualities of Research; Research applications in functional areas of Business, Emerging trends in Business research., Research & the Scientific Method: Characteristics of scientific method. Steps in Research Process

Concept of Scientific Enquiry: – Formulation of Research Problem – Management Question – research Question – Investigation Question

Research Proposal – Elements of a Research Proposal, Drafting a Research Proposal, evaluating a research proposal.

Unit 2 (8 Sessions)

Research design: Concept, Features of a good research design, Use of a good research design; Qualitative and Quantitative research approaches, Comparison – Pros and Cons of both approaches.

Exploratory Research Design: Concept, Types: Qualitative techniques – Projective Techniques, Depth Interview, Experience Survey, Focus Groups, Observation.

Descriptive Research Designs: Concept, types and uses. Concept of Cross-sectional and Longitudinal Research, Experimental Design: Concept of Cause, Causal relationships, Concept of Independent & Dependent variables, concomitant variable, extraneous variable, Treatment, Control group.

Unit 3 (6 Sessions)

Scaling & measurement techniques: Concept of Measurement: Need of Measurement; Problems in measurement in management research – Validity and Reliability. Levels of measurement – Nominal, Ordinal, Interval, Ratio. Attitude Scaling Techniques: Concept of Scale – Rating Scales viz. Likert Scales, Semantic Differential Scales, Constant Sum Scales, Graphic Rating Scales – Ranking Scales – Paired comparison & Forced Ranking – Concept and Application.

Unit 4 (6 Sessions)

Sampling:Basic Concepts: Defining the Universe, Concepts of Statistical Population, Sample, Characteristics of a good sample. Sampling Frame (practical approach for determining the sample frame expected), Sampling errors, Non Sampling errors, Methods to reduce the errors, Sample Size constraints, Non Response.

Probability Sample: Simple Random Sample, Systematic Sample, Stratified Random Sample, Area Sampling & Cluster Sampling., Non Probability Sample: Judgment Sampling, Convenience

Sampling, Purposive Sampling, Quota Sampling & Snowballing Sampling methods. Determining size of the sample – Practical considerations in sampling and sample size, sample size determination.

Unit 5 (8 Sessions)

Data Analysis: Editing, Coding, Tabular representation of data, frequency tables, Construction of frequency distributions, Graphical Representation of Data: Appropriate Usage of Bar charts, Pie charts, Histogram.

Hypothesis: Qualities of a good Hypothesis –Framing Null Hypothesis & Alternative Hypothesis. Concept of Hypothesis Testing – Logic & Importance. Analysis of Variance: One way and two way Classifications.

Mechanism of Report Writing- Report Preparation: Types, Report Structure: preliminary section, main report, interpretation of results, suggestions and recommendations, limitations of the study, Report formulation.

COURSE OUTCOME

Course Outcomes	Blooms Taxanomy
CO1. Knowledge of concept / fundamentals for different types of research.	• Knowledge (K 2)
CO2. Applying relevant research techniques.	• Remembering (K1) • Applying (K 4)
CO3. Understanding relevant scaling & measurement techniques and should use appropriate sampling techniques	• Comprehending (K 3) • Applying (K 4)
CO4.Synthesizing different techniques of coding, editing, tabulation and analysis in doing research.	• Analyzing (K 5) • Synthesizing (K6)
CO5.Evaluating statistical analysis which includes ANOVA technique and prepare research report.	• Evaluating (K7)

Suggested Readings

1. Research Methodology, Deepak Chawla, NeenaSondhi, Vikas Publication
2. Business Research Methods, Naval Bajpai, Pearson Education
3. Research Methodology, C R Kothari, New Age International.
4. Business Research Methods by Donald Cooper & Pamela Schindler, TMGH, 9th Edition.
5. Business Research Methods by Alan Bryman & Emma Bell, Oxford University Press, 2ndEdition.
6. Business Research Methods by T N Srivastava & Shailaja Rao, TMH Publication, 2ndEdition.

FINANCIAL MANAGEMENT AND CORPORATE FINANCE

Course Credit: 3 Contact Hours: 40 Hrs

Course Objectives: This course is intended to introduce the basic theory, concepts and practical applications in corporate finance and to enable students to analyse various corporate decisions. The course objectives are outlined below:

1) To understand the fundamentals, various models and agency problems of Corporate Finance.
2) To acquire knowledge about various techniques used for analysing various long-term projects.
3) To have an understanding about various capital structure techniques and selecting best source of finance.
4) To have an understanding of various dividend models and its applicability.
5) To acquaint students about corporate valuation in mergers and acquisitions.

UNIT I (6 Hrs)

Introduction to Finance & Corporate Finance: Corporate Finance & its scope, Corporate Governance and Agency Problem, Corporate valuation Models: Asset Based Valuation Model, Earning based Valuation Model, Cash flow-based Model, CAPM Model, APT, EVA Analysis, Introduction to start-up finance, Financial Decisions, Time Value of Money.

UNIT II (10 Hrs)

Investment and Financing Decision: Concept of Opportunity Cost, Cost of Debenture, Preference and Equity capital, Composite Cost of Capital, Cash Flows as Profit and components of Cash Flows, Capital Budgeting Decisions, Calculation of NPV and IRR, Excel Application in Analysing Projects.

UNIT III (10 Hrs)

Financial Decision: Capital Structure, Relevance and Irrelevancy theory, Leverage analysis – financial, operating and combined leverage along with its implications, EBIT EPS Analysis, Point of Indifference.

UNIT IV (10 Hrs)

Dividend Relevance: Factors affecting Dividend Policy, Forms of Dividends, Types of Dividend Policies, Dividend Models: Walter and Gordon Model, Miller- Modigliani (MM) Hypothesis.

UNIT V (4 Hrs)

Mergers and Acquisition: Introduction, Exchange Ratio, Synergy Benefits, Post Merger EPS, Post Merger Price of share, Required rate of return of merged company, De-Merger.

Course Outcome: After successful completion of this course students will be able:

S.No	Course Outcome	Bloom's Taxonomy
1.	CO1 Understand the different basic concept / Models of Corporate Finance and Governance	• Knowledge (K2) • Remembering(K1)
2.	CO2 Understand the practical application of time value of money and evaluating long term investment decisions	• Analyzing (K5) • Evaluating(K7)
3.	CO3 Develop analytical skills to select the best source of capital, structure and leverage.	• Analyzing(K5) • Synthesizing(K6)
4.	CO4 Understand the use and application of different models for firm's optimum dividend pay-out.	• Comprehending(K3) • Applying(K4)
5.	CO5 Understand the recent trends of mergers and acquisition and its valuation	• Comprehending(K3) • Synthesizing (K6)

Suggested Readings

1) Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
2) Pandey I M - Financial Management (Vikas, 11th Ed.)
3) William HakkaBettnerCarcello- Financial and Management Accounting (TMH-16th Ed.)
4) Sheebakapil-Fundamental of financial management (Wiley,2015)
5) Prasanna Chandra - Fundamentals of Financial Management (TMH, 9th Ed.)
6) Bark DemazoThampy- Financial Management (Pearson,2nd Ed.)
7) R P Rustagi - Financial Management (Galgotia, 2000, 2nd revised ed.)
8) Damodaran, A., Applied Corporate Finance, 3rd Edition, Wiley, 2012
9) Ravi.M Kishore – Financial Management (Taxman, 7th Ed)
10) Fundamentals to Financial Management, Brigham & Houston, 14/e, Cengage Learning
11) Van Horne - Financial Management and Policy (Prentice hall, 2003, 12th Ed.)

QUANTITATIVE TECHNIQUES FOR MANAGER

Credit 3 Contact Hour: 40

Course Objectives

1. Understand the importance of the use of OR application in decision Making environment
2. To formulate LPP and Obtain Graphical Solutions & Acquire General idea of the Simplex method.
3. To understand and solve transportation & assignment models.
4. To know optimal sequence model and understand concepts of queuing theory.
5. To identify right time for replacement of equipment and understand project management techniques

Unit I (6 Sessions)-Operations Research & Decision Making Environments

Operations Research:- Uses, Scope and Applications of Operation Research in managerial decision-making .*Decision-making environments:-* Decision-making under certainty, uncertainty and risk situations; Decision tree approach and its applications.

Unit II (10 Sessions)-Linear Programming Problem & Transportation Problem

Linear programming: Mathematical formulations of LP Models for product-mix problems; graphical and simplex method of solving LP problems; duality.

Transportation problem: Various methods of finding Initial basic feasible solution-North West Corner Method, Least Cost Method & VAM Method and optimal solution-Stepping Stone & MODI Method, Maximization Transportation Problem

Unit III (8 Sessions)-Assignment model & Game Theory

Assignment model: Hungarian Algorithm and its applications, Maximization Assignment Problem.

Game Theory: Concept of game; Two-person zero-sum game; Pure and Mixed Strategy Games; Saddle Point; Odds Method; Dominance Method and Graphical Method for solving Mixed Strategy Game.

Unit IV (6 Sessions)-Sequencing & Queuing Theory

Sequencing Problem: Johnsons Algorithm for n Jobs and Two machines, n Jobs and Three Machines, Two jobs and m - Machines Problems.

Queuing Theory: Characteristics of M/M/I Queue model; Application of Poisson and Exponential distribution in estimating arrival rate and service rate; Applications of Queue model for better service to the customers.

Unit V (6 Sessions)-Replacement Problem & Project Management

Replacement Problem: Replacement of assets that deteriorate with time, replacement of assets which fail suddenly.

Project Management: Rules for drawing the network diagram, Applications of CPM and PERT techniques in Project planning and control; crashing of operations.

CO1	Be able to understand the characteristics of different types of decision-making environments and the appropriate decision making approaches and tools to be used in each type.	Knowledge (K2)/ Remembering (K1)
CO2	To formulate linear programming problem and to find optimal solution by graphical simplex method.	Knowledge (K2)
CO3	Be able to build and solve Transportation Models and Assignment Models also to solve game theory problems by understanding pure and mix strategies.	Applying (K 4)
CO4	To assign optimal sequence of difference jobs on different machines and develop understanding of queuing theory concepts.	Applying (K 4)
CO5	To implement replacement of equipments at right time and able to implement project management concepts like CPM, PERT to reduce cost and time.	Synthesizing (K6)/ Evaluating (K7)

Suggested Readings

1. R. Panneerselvam - Operations Research (PHI, 2nd Edition)
2. Sharma J K - Operations Research (Pearson, 3rd Edition
3. Apte-Operation Research and Quantitative Techniques (Excel Books)
4. S Kalawathy-Operation Research (Vikas IVth Edition)
5. Natarajan- Operation Research(Pearson)
6. Singh & Kumar—Operation Research(UDH Publisher edition 2013)
7. Taha Hamdy - Operations Research - An Introduction (Prentice-Hall, 9th edition)
8. Vohra - Quantitative Techniques in Management (Tata McGraw-Hill, 2nd)
9. Kothari - Quantitative Techniques (Vikas 1996, 3rd Edition).

OPERATIONS AND GREEN SUPPLY CHAIN MANAGEMEMENT

Course Credit: 3 Contact Hours:40

Course Objectives: -

1. To understand the role of Operations in overall Business Strategy of the firm.
2. To understand the application of operations management policies and techniques to the service sector as well as manufacturing firms.
3. To identify and evaluate the key factors and their interdependence of these factors in the design of effective operating systems.
4. To understand the importance of Green Logistics and Supply Chain management
5. To familiarize the students with the techniques for effective utilization of operational resources and managing the processes to produce good quality products and services at competitive prices.

UNIT –I (6 sessions) Production Concepts:

Introduction, meaning, nature and scope of production and operations management. Difference between production and operations management. Productivity, factors affecting productivity and productivity measurement. Work study— Method study and work measurement. Production Technology – Types of manufacturing processes. Plant location and types of plant layout.

UNIT –II (6 sessions) Operations Concepts:

Services scenario in India, difference between product and service, characteristics of services, classification of services, product and service design, factors affecting service design, service designing process, service blueprinting, service capacity planning. Dimensions of quality in services, understanding service quality gap, measuring service quality using SERVQUAL model.

UNIT-III (10 sessions) Material and Inventory Management:

Types of production planning, process of production planning and control (PPC) – routing, scheduling and loading. Master production schedule, aggregate production planning. Types of inventories, inventory control techniques- EOQ, ABC, VED, FSN, HML and SDE (Simple numerical problems on Inventory control techniques). Just-in-time (JIT) and KANBAN.

UNIT-IV (10 sessions) Green Supply Chain Management:

Introduction – Traditional Supply Chain and Green Supply Chain – Environmental Concern and Supply Chain – Closed-loop Supply Chain – Corporate Environmental Management – Green Supply Chain (GSCM): Definition, Basic Concepts – GSCM Practices. Case Studies

UNIT-V (8 sessions) Green Logistics

Green Logistics and Transportation – Definitions of Green Logistics – Critical drivers of Green Logistics – Green transportation and logistics practices – Environmental impacts of transportation and logistics – Closing the Loop: Reverse Logistics. Case Studies

Suggested Readings:-

1. Aswathappa, K. & Bhat, K.S.-- Production and Operations Management (Himalaya Publishing House, 2nd Edition)
2. Chase, R.B., Shankar, R. & Jacobs, F.R. -- Operations & Supply Chain Management (Tata McGraw Hill, 14th Edition)
3. Chunawalla, S.A. & Patel, D.R. – Production & Operations Management (Himalaya Publishing House, 9th Edition)
4. Chary, S.N. - Production and Operations Management (Tata McGraw Hill, 6th Edition)

5. Joseph Sarkis, Yijie Dou. Green Supply Chain Management: A Concise Introduction, Routledge, 2017.
6. Charisios Achillas, Dionysis D. Bochtis, Dimitrios Aidonis, Dimitris Folinas. Green Supply Chain Management, Routledge, 2018.
7. Charantimath, P.M. – Total Quality Management (Pearson Education, 3rd Edition)
8. Bedi, Kanishka – Production & Operations Management (Oxford University Press, 3rd Edition)
9. Adam, Everett E. & Ebert, Ronald J. – Production and Operations Management (Prentice Hall, 5th Edition)
10. Gopalakrishnan, P. & Sundaresan, M. – Materials Management (Prentice Hall of India)

PROCUTEMENT, STORAGE AND WAREHOUSE MANAGEMENT

Course Credit: 3 **Contact Hours:40**

Course Objectives: -

1. To provide understanding of Procurement Process
2. To provide understanding of Storage and Warehouse management
3. To enable the critical thinking for optimum utilizations of storage system
4. To use the latest approach in warehouse management
5. To be familiar with laws and safety regulations in warehouse management

Unit I (6 Hours)

Objectives of Procurement System, Principles of Procurement, History of procurement function: from administrative to strategic, value added role, Procurement Cycle, Procurement Planning, Purchasing Mix: Six Rights, Selecting the right supplier, Source of information and process, Supplier appraisal/vendor capability, Bidding process. **Case Studies**

Unit II (9 Hours)

Storage Management system – Storage Inventory Management – Functions of storage & Inventory - Classification of Inventory- Methods of Controlling Stock Levels- Always Better Control (ABC) Inventory system- Storage: storage policies - dedicated storage, randomized storage & class-based storage; Storage Methods-assembling & seasonal storage; stockpiling and rapid storage. Centralized and Decentralized Storage Systems. **Case Studies**

Unit III (9 Hours)

Introduction to Warehousing: Evolution of warehousing from store to warehouse, warehouse operations, process of receiving and put away, principle of storing goods & various storing methods, process of order picking and order creation, significance of packaging, documents required for issuing goods. **Case Studies**

Unit IV (8 Hours)

Strategic Warehousing: Meaning & benefits of Strategic Ware Housing, Types of Warehouses; Warehouse Design: Elements & Principle of Warehouse Design, factors of warehouse design; Warehouse Location: its benefits & Significance of Warehouse in SCM. **Case Studies**

UNIT V (8 Hours)

Warehousing Operations: - inbound process, outbound processes, Functions of Warehouse- break-bulk, cross docking, order mixing. MHEs in warehouse, legal requirements for ensuring a safe workplace; and Warehouse Management Systems. **Case Studies**

Suggested Readings

1. GWYNNE RICHARDS (2014) Warehouse Management: A Complete Guide to Improve Efficiency and Minimizing Cost in the Modern Warehouse. The Chartered Institute of Logistics and Transport, Kegan page limited.
2. DAVID E. MULCHY & JOACHIM SIDON (2008) A Supply Chain Logistics Program for Warehouse Management. Auerbachian Publications References

1. 3.. Bowersox, D.J., Closs, D.J., Cooper, M.B., & Bowersox, J.C. (2013). Supply Chain Logistics Management. (4 th ed.), McGraw Hill/Irwin.
3. Arnold, J.R., Chapman, S.N. (2012). The Introduction to Materials Management. (7 th ed.), Prentice-Hall.
2. Coyle, J.J., Jr. Langley, C.J., Novack, R.A, & Gibson, B.J. (2013). Managing Supply Chains: A Logistics Approach. (9 th ed.), McGrawHill.
3. Edward, F. (2002). 3. World-Class Warehousing and Material Handling. (International ed.), McGraw-Hill. Muller, M. (2011). Essentials of Inventory Management. (2 nd ed.), American Management Association.

PORT AND AIRPORT MANAGEMENT FOR LOGISTICS

Course Credit: 3 **Contact Hours:40**

Course Objectives:

1. To be familiar with working of Ports and Airports
2. To understand the operating procedures of logistics at Port and Airport
3. To be familiar with norms of logistics at port and airport.
4. To be aware of the development of facilities at port in the era of globalization
5. To be able to analyses the performance of standard operating procedure at port and airport for cargo

Unit 1 (8 Hours)

Port Structure and Functions: Definition - Types and Layout of the Ports – Organisational structure- Fundamental observations. Main functions and features of ports: Infrastructure and connectivity Administrative functions - Operational functions. Main services: Services and facilities for ships - Administrative formalities - Cargo transfer - Services and facilities for cargo - Additional "added value" service- Ports and their stakeholders like PHO, Immigration, Ship agents, Stevedores, CHA

Unit II(8 Hours)

Port Operations: Berths and Terminals - Berth Facilities and Equipment - ship Operation – Pre shipment planning, the stowage plan and on-board stowage - cargo positioning and stowage on the terminal - Developments in cargo/container handling and terminal operation - Safety of cargo operations - Cargo security: Measuring and evaluating performance and productivity.

Unit III(8 Hours)

Port Development: Phases of port development - Growth in word trade - Changes in growth Development in terminal operation. Shipping technology and port: Ship knowledge Ship development and port development - Port time and ship speed - Other technical development affecting port.

UNIT IV(8 Hours)

Port Administration Ownership and Management Port ownership structure- Types of port ownership and administration – Organizations concerning ports - Boards governing the ports - Port management development Rise and fall of Ports - information technology in ports. Port ownership in Indian context: Acts governing the Ports in India - Port ownership structure in India. Port reform: Framework for port reform - Evolution of ports in a competitive world Alternative Port Management Structure and Ownership Models.

UNIT V(8 Hours)

Air Transport: Introduction to Air Transport – Air Freight – IATA – Cargo Handling at Goods at Air Port – Information Management of Air Cargo – System and Modules – Distribution of Goods.

Suggested Readings

1. PATRICK M.ALDERTON. 2008, Port Management and Operations. Information Law Category, U.K. Reference Books
2. WORLD BANK. 2007, Port Reform Tool Kit. World Bank, Washington.
3. MARIA G.BURNS. 2014., Port Management and Operations. CRS Press, U.K.
4. ALAN E.BRANCH. 2008, Elements of Shipping. Chapman and Hall, Fairplay Publications, U.K.
5. DE MONIE. 1989., Measuring and Evaluating Port Performance and Productivity. UNCTAD, New York.

IT SKILLS LAB-2

Lab work 20 Hours

Course Objective

1. To develop pivot table and understand the validating & auditing techniques
2. To understand different charting techniques in MS Excel
3. To understand different formatting techniques in MS Excel

Unit I (Lab work on spreadsheet)

Pivot Table: Developing Pivot Table, Analyzing data using goal seek and solver, Scenarios Create named scenarios. Show, edit, delete scenarios, Creating a scenario summary report. Validating and Auditing: Set, edit validation criteria for data entry in a cell range like: whole number, decimal, list, date, time, Trace precedent, dependent cells. Identify cells with missing dependents. Creating applications in Spreadsheet and Macros.

Unit II (Lab work on spreadsheet) 15 Hours

Creating and formatting Charts: Understanding chart types, column chart, bar chart, line chart, pie chart, XY Scatter chart , Area chart, surface chart, bubble chart. Create a combined chart like: column and line, column and area. Change the chart type for a defined data series, Add, delete a data series in a chart, Re-position chart title, legend, data labels. Change scale of value axis: minimum, maximum number to display, major interval. Change display units on value axis without changing data source: hundreds, thousands, millions. Format columns, bars, pie slices, plot area, chart area to display an image.

References

Excel Data Analysis: Modeling and Simulation , Hector Guerrero (Springer)

COURSE OUTCOME

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. To gain knowledge of pivot table and understand the validating & auditing techniques	Knowledge (K2)
2	CO2. Learn to use different charting techniques in MS Excel	Applying (K4) Synthesizing (K6)
3	CO3. Learn to use different formatting techniques in MS Excel	Applying (K4) Knowledge (K2)

MINI PROJECT -2

Course Credit: 2

Seminar by students

Objective –

1. To identify the issues challenge of the industry
2. To able to prepare report on the application of emerging technologies in the selected industry

In second semester, the students are required to take one industry as per his/her interest for analysis and preparing a project report. Preference should be given on the application of emerging technologies in the selected industry. It may consists of Fintech, Block chain, Financial Services, Data Science, Social Entrepreneurship or any other suitable area of interest. The report will be prepared individually. The report will be evaluated by one external examiner appointed by university.

COURSE OUTCOME

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. To gain knowledge of issues challenge of the industry	Knowledge (K2)
2	CO2. Learn to prepare report on the application of emerging technologies in the selected industry	Applying (K4) Synthesizing (K6)